

Software Engineering Assignment: SDLC Model Selection

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Course: Software Engineering

Topic: SDLC Model Selection

Objective: To analyse different software project scenarios and select the most suitable SDLC model with proper justification

1. Introduction

A Library Management System (LMS) is a software application designed to automate and manage the core operations of a library. It replaces manual record-keeping with a centralized digital system, improving accuracy, efficiency, and accessibility.

The system operates on a database-driven architecture, where all records (books, users, transactions) are stored and retrieved using structured queries.

Core Functional Modules:

- Book Management: Handles operations such as adding new books, updating book details, deleting records, and tracking availability. Each book is uniquely identified using attributes like ISBN, title, author, and category.
- User Management: Maintains user profiles (students, staff, admin) with authentication mechanisms. It may include role-based access control (RBAC) to restrict functionalities.
- Issue/Return System: Manages book transactions. It records issue dates, due dates, and return status, ensuring transaction consistency.
- Fine Calculation Module: Automatically calculates fines based on late returns using predefined rules (e.g., fine per day). This involves basic business logic implementation.
- Search System: Enables efficient querying using filters like title, author, or category. Internally, it may use indexing techniques for faster retrieval.

2. Selected SDLC Model

Waterfall Model

3. Justification

1. Requirement Stability

- The requirements of an LMS are clearly defined and rarely change over time. Functions such as issuing books, returning books, and maintaining user records are standard across most libraries.
- From a software engineering perspective:
- Requirements can be documented in a Software Requirement Specification (SRS) at the beginning
- No need for frequent requirement re-evaluation

- The Waterfall model is ideal when requirements are predictable and frozen early.

2. Risk Level

- The system has a low technical and operational risk:
- No real-time external integrations (e.g., payment gateways, third-party APIs)
- No high-security financial transactions
- Limited concurrency issues
- Thus:
- Risk analysis is straightforward
- Failure impact is minimal
- Waterfall is suitable for low-risk environments where extensive risk iteration is unnecessary.

3. Complexity

- The LMS is moderately simple in terms of system design:
- Primarily based on CRUD operations
- Uses a relational database model (tables like Books, Users, Transactions)
- Limited business logic
- From an architectural perspective:
- Can be implemented using a 3-tier architecture:
- Presentation Layer (UI)
- Application Layer (logic)
- Data Layer (database)
- Since complexity is manageable, a linear development model like Waterfall is sufficient.

4. User Involvement

- User interaction during development is minimal and mostly limited to initial requirement gathering.
- Librarians already understand workflows
- No need for iterative feedback cycles
- System behaviour is predictable
- Waterfall works well when continuous user feedback is not critical.

5. Flexibility

- The system does not demand high flexibility or adaptability:
- Features are predefined and standardized
- Changes (if any) are minor and infrequent
- From a maintenance perspective:
- Updates can be handled in the maintenance phase without affecting earlier stages
- Waterfall is appropriate when flexibility is not a primary concern.

4. Comparison with Other Models

- Agile Model
- The Agile model emphasizes iterative development, continuous feedback, and rapid changes.
- However, for LMS:
- Requirements are stable → iteration is unnecessary
- Frequent releases are not required
- Continuous testing cycles increase overhead

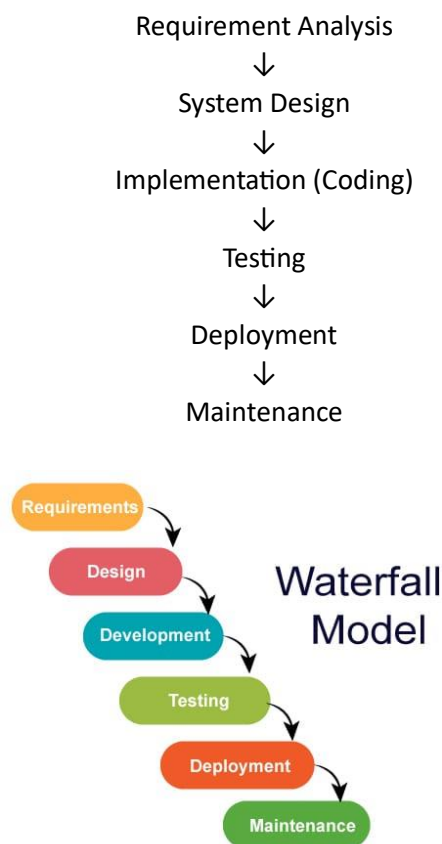
- Agile introduces unnecessary complexity for a simple, stable system.

Spiral Model

- The Spiral model focuses on risk assessment and iterative refinement.
- For LMS:
- Risk factors are minimal
- No need for repeated risk analysis cycles
- Higher cost due to complexity
- Spiral is over-engineered for this type of application.

5. Diagram

Waterfall Model Diagram:



6. Conclusion

The Waterfall Model is the most suitable SDLC model for a Library Management System due to its stable requirements, low risk, and straightforward architecture. Its structured and sequential approach ensures clear documentation, predictable development, and efficient implementation, making it ideal for such systems.